Furmonertinib (AST2818) versus gefitinib as first-line therapy for Chinese patients with locally advanced or metastatic *EGFR* mutation-positive non-small-cell lung cancer (FURLONG): a multicentre, double-blind, randomised phase 3 study

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Summary

Background Furmonertinib (AST2818) is an irreversible, selective, third-generation EGFR tyrosine-kinase inhibitor. We aimed to investigate the efficacy and safety of furmonertinib versus the first-generation EGFR tyrosine-kinase inhibitor gefitinib as first-line treatment in patients with *EGFR* mutation-positive locally advanced or metastatic non-small-cell lung cancer (NSCLC).

Methods The FURLONG study is a multicentre, double-blind, randomised, phase 3 study done in 55 hospitals across mainland China. We enrolled patients who were aged 18 years or older and had histologically confirmed, locally advanced or metastatic, stage IIIB, IIIC, or IV unresectable NSCLC with *EGFR* exon 19 deletions or exon 21 Leu858Arg mutation on tissue biopsy confirmed by a central laboratory. Eligible patients were stratified according to *EGFR* mutation (exon 19 deletions or exon 21 Leu858Arg) and CNS metastases (with or without) and randomly assigned (1:1) to receive either oral furmonertinib (80 mg/day) or oral gefitinib (250 mg/day) in 21-day cycles until disease progression, the occurrence of intolerable toxicities, withdrawal of consent, or other discontinuation reasons judged by the investigators. Investigators, clinicians, participants, independent review centre (IRC) members, the sponsor, and those analysing the data were all masked to treatment allocation. The primary endpoint was IRC-assessed progression-free survival and, along with safety, was analysed in the full analysis set, which comprised all randomly assigned patients who had received at least one dose of study drug. This study is registered with ClinicalTrials.gov, NCT03787992, and is ongoing for survival follow-up.

Findings Between May 30, 2019, and Dec 5, 2019, 750 patients were screened, of whom 358 were randomly assigned to receive either furmonertinib and gefitinib-matching placebo (n=178) or gefitinib and furmonertinib-matching placebo (n=180). 178 patients randomly assigned to furmonertinib and 179 patients randomly assigned to gefitinib were treated and were included in the full analysis set. Median follow-up was 21·0 months (IQR 18·0–23·5) in the furmonertinib group and 21·0 months (18·0–23·5) in the gefitinib group. Median IRC-assessed progression-free survival was 20·8 months (95% CI 17·8–23·5) in the furmonertinib group and 11·1 months (9·7–12·5) in the gefitinib group (hazard ratio 0·44, 95% CI 0·34–0·58; p<0·0001). Treatment-related adverse events of a grade 3 or more occurred in 20 (11%) of 178 patients in the furmonertinib group and in 32 (18%) of 179 patients in the gefitinib group. Treatment-related serious adverse events were reported in ten (6%) patients in the furmonertinib group and in 11 (6%) patients in the gefitinib group. Ten (6%) patients in the furmonertinib group and three (2%) patients in the gefitinib group died due to adverse events, which were all judged to be possibly unrelated to study treatment by the investigators.

Interpretation Furmonertinib showed superior efficacy compared with gefitinib as first-line therapy in Chinese patients with *EGFR* mutation-positive NSCLC, along with an acceptable safety profile without new signals. Furmonertinib is a new potential treatment option for this population.

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For the Chinese translation of the abstract see Online for appendix 1

*A list of the FURLONG investigators can be found in appendix 2 (pp 3–6)

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Introduction

EGFR is one of the most common driver oncogenes detected in non-small-cell lung cancer (NSCLC) of adenocarcinoma histology, with a prevalence of 10% in White patients,[1] 51·4% in Asian patients, and 50·2% in Chinese patients.[2] EGFR tyrosine-kinase inhibitors are the first treatment choice in patients with NSCLC with *EGFR* sensitising mutations because they have shown

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See Online for appendix 2

Research in context

Evidence before this study

Before planning the FURLONG study, first-generation and second-generation EGFR tyrosine-kinase inhibitors were standard first-line treatments for patients with *EGFR* mutation-positive non-small-cell lung cancer (NSCLC) in mainland China. Most studies report that progression-free survival for patients with *EGFR*-mutated NSCLC treated with first-generation or second-generation EGFR tyrosine-kinase inhibitors as the first line treatment is around 10 months. Adverse events, especially those associated with drug action on wild-type *EGFR* (eg, diarrhoea and rash), are common with these first-generation and second-generation drugs. We searched PubMed without language restrictions using the string of "(lung cancer) AND (*EGFR*)" and limited the results to phase 2 and phase 3 studies published between Jan 1, 2016, and Dec 8, 2021. To the best of our knowledge, osimertinib is the only third-generation EGFR tyrosine-kinase inhibitor approved for first-line therapy of *EGFR* mutation-positive NSCLC, with a published phase 3 study (FLAURA) comparing osimertinib with first-generation EGFR tyrosine-kinase inhibitors.

Added value of this study

This study shows that furmonertinib was superior to gefitinib in terms of progression-free survival as first-line treatment for Chinese patients with *EGFR* mutation-positive NSCLC. This benefit was consistent in most prespecified subgroups, including in patients with CNS metastases. Patients in the furmonertinib group had a lower prevalence of treatment-related adverse events of a grade 3 or more, diarrhoea, rash, and elevated aminotransferase enzymes than did patients in the gefitinib group.

Implications of all the available evidence

Irrespective of the presence of CNS metastases, patients with *EGFR* mutation-positive NSCLC can significantly benefit from receiving furmonertinib over gefitinib as first-line therapy. Furmonertinib also showed an acceptable safety profile without new signals. Furmonertinib is a new potential first-line therapy for Chinese patients with *EGFR*-mutated NSCLC.

good efficacy in well designed randomised controlled trials.[3–6] However, disease progression inevitably occurs and median progression-free survival in patients with *EGFR*-mutated NSCLC remains about 10 months.[3–6]

In one cohort, the Thr790Met *EGFR* mutation was detected in 98 (63%) of 155 patients with NSCLC who had disease progression on first-generation EGFR tyrosine-kinase inhibitors.[7] Resistance to EGFR tyrosine-kinase inhibition mediated by the Thr790Met mutation has led to the development of third-generation EGFR tyrosine-kinase inhibitors such as osimertinib.[7] Besides overcoming the Thr790Met mutation, osimertinib also showed antitumour activity against *EGFR* sensitising mutations (deletions within exon 19 and Leu858Arg in exon 21) in vitro and had lower activity against wild-type *EGFR* cell lines than against mutant *EGFR* cell lines.[8] In the FLAURA study, osimertinib was superior to gefitinib or erlotinib in terms of progression-free survival (18·9 months [95% CI 15·2–21·4] *vs* 10·2 months [9·6–11·1]; p<0·001) and overall survival (38·6 months [34·5–41·8] *vs* 31·8 months [26·6–36·0]; p=0·046) as first-line therapy for *EGFR* sensitising mutation-positive NSCLC.[9] Osimertinib also reduces the risk of CNS progression compared with first-generation EGFR tyrosine-kinase inhibitors in patients with *EGFR*-mutated NSCLC.[10] On the basis of the results of the FLAURA study, osimertinib has become an approved first-line therapy in *EGFR*-mutated NSCLC.

Furmonertinib (AST2818) is a novel third-generation EGFR tyrosine-kinase inhibitor that targets both *EGFR* sensitising mutations and the Thr790Met mutation while sparing wild-type *EGFR*.[11] Furmonertinib has shown encouraging efficacy and good tolerability in previous phase 1/2 and phase 2b studies of patients with *EGFR* Thr790Met-mutated NSCLC.[11,12] Furmonertinib showed activity in the CNS lesions of patients with *EGFR*-mutated NSCLC, with an objective response rate of 66% (95% CI 46–82).[12] Adverse events related to drug action on wild-type *EGFR*, such as diarrhoea and rashes, were infrequent in these studies.[11,12] Here, we aimed to compare furmonertinib with gefitinib as first-line treatments for patients with locally advanced or metastatic *EGFR*-mutated NSCLC.

Methods

Study design and participants

The FURLONG study is a multicentre, double-blind, randomised, phase 3 study done in 55 hospitals across mainland China. Eligible patients were aged at least 18 years and had histologically confirmed, unresectable, locally advanced or metastatic NSCLC (eg, could be systemic recurrence or newly diagnosed stage IIIB, IIIC, or IV based on the eighth version of the American Joint Committee on Cancer lung cancer staging system). Patients had an Eastern Cooperative Oncology Group (ECOG) performance status of 0–1, *EGFR* mutations (either exon 19 deletions or exon 21 Leu858Arg) confirmed by a central laboratory through tissue biopsy by use of the cobas *EGFR* Mutation Test (version 2; Roche Molecular Systems, Pleasanton, CA, USA), and at least one target lesion that had not been locally treated and was measurable according to Response Evaluation Criteria in Solid Tumours (RECIST) version 1.1. Additionally, patients' health could not deteriorate in the 2 weeks before ECOG scoring and patients had a life expectancy of at least 12 weeks. Patients with

asymptomatic CNS metastases that did not require steroid treatment for 28 days or more could also be enrolled. A brain scan via CT or MRI was mandatory for all patients before enrolment.

Key exclusion criteria were the receival of previous systemic antitumour therapies for advanced or metastatic NSCLC, including EGFR tyrosine-kinase inhibitors; surgery or large area radiotherapy within 28 days before the first dose of study drug; local or palliative radiotherapy for bone metastases or other clinical investigational products within 14 days before the first dose of study drug; strong cytochrome P450 3A4 inducers or inhibitors or traditional Chinese medicines with antitumour indications within 7 days before the first dose of study drug; and medicines that can prolong the heart-rate corrected QT (QTc) interval on electrocardiogram (ECG). Patients with toxicities related to previous antitumour therapies that did not recover to grade 1 according to the Common Terminology Criteria for Adverse Events (CTCAE, version 5.0; neuropathy or alopecia induced by chemotherapy could be grade 2); a mixed histological type of lung cancer; another malignant tumour or a history of other malignant tumours; severe gastrointestinal disorders that might affect drug absorption; inadequate organ function; a mean QTc interval of more than 470 ms on ECG; clinically significant ECG abnormalities; any risk factors for prolonged ECG QTc interval or arrhythmia; a left ventricular ejection fraction of less than 50%; active infectious disease (ie, hepatitis C virus, hepatitis B virus, or HIV); a history of interstitial lung disease or its risk factors; or other conditions unsuitable for the study based on investigator judgement were excluded.

The protocol received ethics approval from the institutional review board or ethics committee of each participating centre. The study was done in accordance with Good Clinical Practice guidelines (as defined by the International Conference on Harmonisation) and the Declaration of Helsinki. Written informed consent was obtained from each patient before enrolment.

Randomisation and masking

Eligible patients were stratified according to *EGFR* mutations (exon 19 deletions or exon 21 Leu858Arg) and CNS metastases (with or without) and were randomly assigned (1:1) to receive either furmonertinib or gefitinib. Randomisation was done by an independent team from a company named Aikeman (Nanjing, China) using a central interactive web response system provided by Medidata (New York, NY, USA), who had no other involvement in this study. Each patient who was randomly assigned had a unique randomisation code generated by the interactive web response system. Because the appearances of furmonertinib (yellow) and gefitinib (brown) are different, a double-dummy method was used to blind the study drug, which was also coded by the interactive web response system. Patients received either furmonertinib plus dummy gefitinib placebo or gefitinib plus dummy furmonertinib placebo. The placebos simulated the colours and appearances of their respective active drugs. Investigators, clinicians, participants, independent review centre (IRC) members, the sponsor, and those analysing the data were all masked to treatment allocation. Unblinding would be done after the primary analysis but could be done beforehand if urgent interventions were required, the *EGFR* Thr790Met mutation was confirmed after progression, or next-line treatment could be affected by blinding status when the patient did not have the *EGFR* Thr790Met mutation after progression. In cases of urgent unblinding, treatment information would not be provided to the sponsor.

Procedures

After enrolment, patients received 80 mg of oral furmonertinib (two tablets taken once daily) plus gefitinib-matching oral placebo once daily or 250 mg of oral gefitinib (one tablet) plus furmonertinib-matching oral placebo once daily in 21-day cycles until disease progression, the occurrence of intolerable toxicities, withdrawal of consent, or other discontinuation reasons judged by the investigators. Adverse events were collected and graded according to CTCAE version 5.0 at every study visit per protocol until the visit after disease progression or discontinuation of study drugs. Dose reductions were permitted for furmonertinib or furmonertinib-matching placebo after a dose interruption of less than 21 days (for diarrhoea and skin disorders, dose interruption was <14 days) due to grade 3 or 4 adverse events. The doses of furmonertinib or furmonertinib-matching placebo could be reduced by half but dose reduction of gefitinib or gefitinib-matching placebo was not allowed according to the prescribing information of gefitinib. After dose reduction, dose escalation was not permitted. If the same adverse events deteriorated to grade 3 or 4 again after dose reduction and did not recover to grade 1 or baseline level after dose interruption for 14 days or 21 days, the study drug would be permanently discontinued.

Tumour assessments by CT or MRI were done at screening, every two cycles in the first 18 cycles, and then every four cycles or when deemed clinically necessary until the visit after disease progression (the final visit for tumour assessment). Patients who discontinued the study drug for reasons other than disease progression were followed up on the same schedule until disease progression, death, or withdrawal of consent, whichever came first. Tumour responses were assessed by a blinded IRC and investigators per RECIST version 1.1. After disease progression, patients were followed up for survival status every 12 weeks until death or loss to follow-up. Physical examination, blood routine examination, blood chemistry, urinalysis, and recording of patient-reported outcomes (European Organisation for Research and Treatment of Cancer

Quality of Life Questionnaire-Core 30 items [EORTC QLQ-C30][13] and the corresponding Lung Cancer 13 module [EORTC QLQ-LC13])[14] were done on day 0 of cycle 1 and day 1 of the following cycles until treatment cessation. Blood routine examination, blood chemistry, and urinalysis were also done on day 8 of cycle 1. *EGFR* Thr790Met mutation status was tested with tissue or blood samples taken after disease progression. Patients with the *EGFR* Thr790Met mutation were unmasked to treatment allocation and next-line therapy was decided by investigators.

Outcomes

The primary endpoint was progression-free survival centrally assessed by an IRC and was defined as the time from randomisation to disease progression or death from any cause, whichever occurred first. Secondary endpoints were progression-free survival assessed by investigators; overall survival (defined as the time from randomisation to death from any cause); objective response rate (the proportion of patients with a complete or partial response), disease control rate (the proportion of patients with a complete response, partial response, or stable disease), duration of response (the time from the date of first documented response to the date of disease progression or death, whichever occurred first), depth of response (best percentage change in the sum of the longest diameter of target lesions compared with baseline in patients without progression of non-target lesions and the occurrence of new lesions), and time to progression (time from randomisation to the earliest occurrence of disease progression) as assessed by investigators and the IRC; the type, frequency, severity, and degree of treatment-emergent adverse events (according to CTCAE version 5.0) and their causal relationship with study drugs, as assessed by investigators; patient-reported outcomes on EORTC QLQ-C30 and EORTC QLQ-LC13; IRC-assessed and investigator-assessed progression-free survival and objective response rate and overall survival in prespecified subgroups (*EGFR* exon 19 deletions and *EGFR* Leu858Arg in exon 21; with or without CNS metastases), all of which (except for progression-free survival in CNS metastases subgroups) will be reported in future publications; clinically significant laboratory test findings; ECG findings; left ventricular ejection fraction; physical examination findings; and CNS progression-free survival, CNS duration of response, CNS objective response rate, and 6-week and 12-week CNS disease control rate, as assessed by the IRC (CNS efficacy analyses will be reported in future publications). All response assessments were reviewed according to RECIST version 1.1.

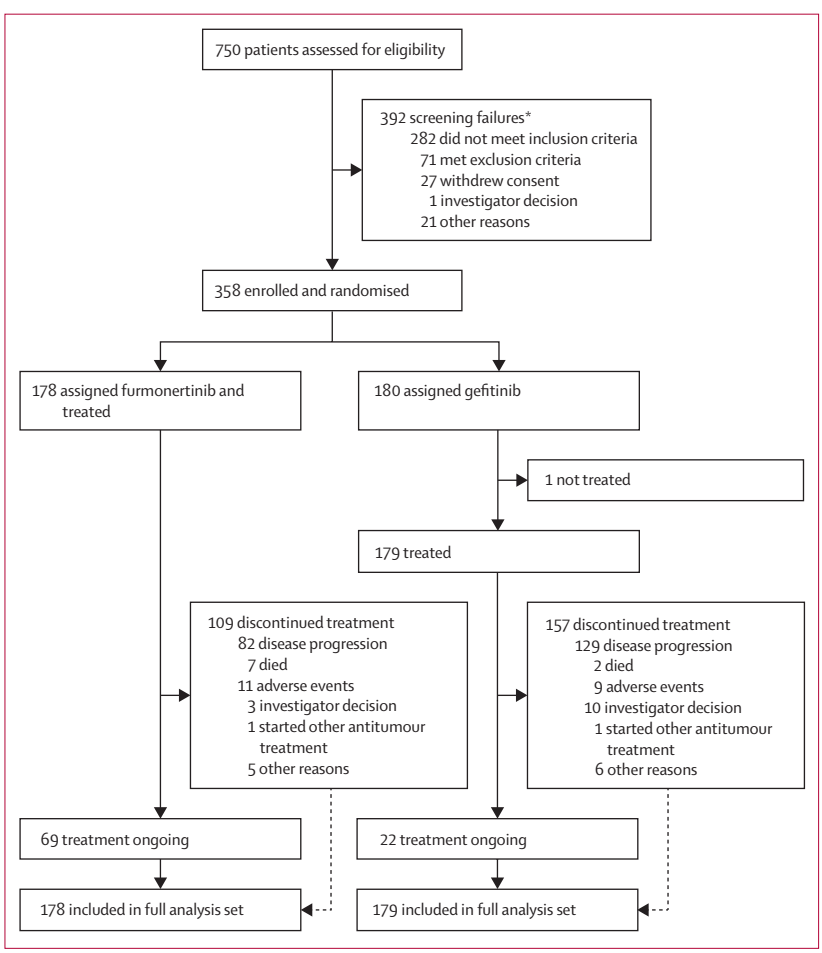

Figure 1: **Trial profile**
*Ten patients had two reasons for screening failure.

Statistical analysis

The primary endpoint of this study was to compare IRC-assessed progression-free survival between the furmonertinib and gefitinib groups. On the basis of published data[15] on first-generation and third-generation EGFR tyrosine-kinase inhibitors, we estimated that progression-free survival would be 16 months in the furmonertinib group and 11 months in the gefitinib group. Assuming a randomisation ratio of 1:1 and an enrolment duration of 6·5 months, 358 patients with a minimum of 238 progression-free survival events were needed to provide at least 80% power to detect a hazard ratio (HR) of 0·688 at a two-sided α level of 0·05. An interim analysis was planned by an independent data monitoring committee after approximately 119 progression-free survival events had occurred. If the results for the primary endpoint were statistically significant at the interim analysis (two-sided p=0·00306), the independent data monitoring committee might recommend stopping the study for efficacy. For further information about protocol amendments and the interim analysis, please refer to appendix 2 (pp 6–7). According to the Lan–DeMets approach, a two-sided p value of 0·049 was considered to be statistically significant for the final analysis of progression-free survival. The primary endpoint was analysed by the cutoff date of Sept 15, 2021.

All efficacy endpoints and patient-reported outcomes were assessed in the full analysis set, which comprised all randomly assigned patients who had received at least one dose of study drug. We also analysed IRC-assessed progression-free survival in a per-protocol set, which comprised all patients who had completed treatment according to the protocol or had no major protocol deviation based on data review. Safety endpoints were assessed in the full analysis set.

Time-to-event endpoints (ie, progression-free survival, overall survival, duration of response, and time to progression) and corresponding p values were analysed with the log-rank test, adjusted with the stratification factors of *EGFR* mutation type (exon 19 deletions or exon 21 Leu858Arg) and CNS metastases status (with or without) at baseline. Median values and two-sided 95% CIs were analysed by use of the Kaplan–Meier method. We calculated HRs and 95% CIs using the Cox proportional hazards model, also adjusting for the stratification factors of *EGFR* mutation type and CNS metastases status. Final overall survival will be analysed at 60% maturity.

We calculated objective response rate and disease control rate on the basis of the best response recorded during the study and the corresponding two-sided 95% CIs using the Clopper–Pearson method. Odds ratios (ORs) and 95% CIs of objective response rate and disease control rate were calculated with logistic regression and p values were calculated with the Wald χ^2 test. Depth of response was compared by use of a covariance analysis. For the analysis of EORTC QLQ-C30 and EORTC QLQ-LC13 scores, a mixed model for repeated measures was used. An increase in functional scales and a decrease in symptom scales indicate improvements in health status. A difference of 10 points or more is of clinical relevance.

We did a prespecified subgroup analysis of IRC-assessed progression-free survival in the full analysis set by sex, age at screening, ECOG performance status, smoking history, *EGFR* mutation, and CNS metastases. For this analysis, HRs and p values were estimated with an unstratified Cox proportional hazards model and an unstratified log-rank test. The subgroup analysis of progression-free survival is for exploratory purposes to evaluate the consistency of the treatment effect in different patient groups. No adjustment to the significance level for testing was made. Post-hoc, we analysed IRC-assessed progression-free survival in patients with concurrent insertions in exon 20 of *EGFR* at baseline. Statistical analyses were done by use of SAS, version 9.2. This study is registered with ClinicalTrials.gov, NCT03787992.

Role of the funding source

The study was designed by the sponsor and the principal investigator (YS). The sponsor provided funding and organisational support and had a role in data collection, data analysis, data interpretation, and writing of the report.

	Furmonertinib (n=178)	Gefitinib (n=179)
Age		
Median, years	59 (53–67)	60 (52–67)
<65 years	118 (66%)	114 (64%)
≥65 years	60 (34%)	65 (36%)
Sex		
Female	116 (65%)	111 (62%)
Male	62 (35%)	68 (38%)
ECOG performance status		
0	39 (22%)	28 (16%)
1	138 (78%)	151 (84%)
2	1 (1%)	0
EGFR mutation subtype		
Exon 19 deletions	91 (51%)	92 (51%)
Exon 21 Leu858Arg	87 (49%)	87 (49%)
History of smoking		
Yes	41 (23%)	44 (25%)
No	137 (77%)	135 (75%)
Disease stage		
III	10 (6%)	7 (4%)
IV	168 (94%)	172 (96%)
CNS metastases*		
Yes	63 (35%)	58 (32%)
No	115 (65%)	121 (68%)

Data are median (IQR) or n (%). All patients were Chinese. ECOG=Eastern Cooperative Oncology Group. *CNS metastases were identified from baseline data for the CNS lesion site and medical history, including previous surgery and radiotherapy for CNS lesions, all of which were reported by investigators.

Table 1: **Baseline demographic and clinical characteristics of patients in the full analysis set**

Results

Between May 30, 2019, and Dec 5, 2019, 750 patients were screened and 358 eligible patients were randomly assigned to receive either furmonertinib (n=178) or gefitinib (n=180; figure 1). Baseline demographic and clinical characteristics were well balanced between the groups (table 1). One patient who was randomly assigned to the gefitinib group did not receive the study drug and was excluded from the full analysis set. 176 patients in the furmonertinib group and 178 patients in the gefitinib group were included in the per-protocol set, as two patients had major protocol violations in the furmonertinib group and one patient had a major protocol violation and one patient did not receive treatment in the gefitinib group (appendix 2 p 8).

An interim analysis was done at 117 progression-free survival events by the data cutoff date of July 1, 2020, by an independent data monitoring committee and the study continued without modification (appendix 2 pp 6–7). By the data cutoff date of Sept 15, 2021, 69 (39%) of 178 patients in the furmonertinib group and 22 (12%) of 179 patients in the gefitinib group remained on study treatment. Median follow-up was 21·0 months (IQR 18·0–23·5) in the furmonertinib group and

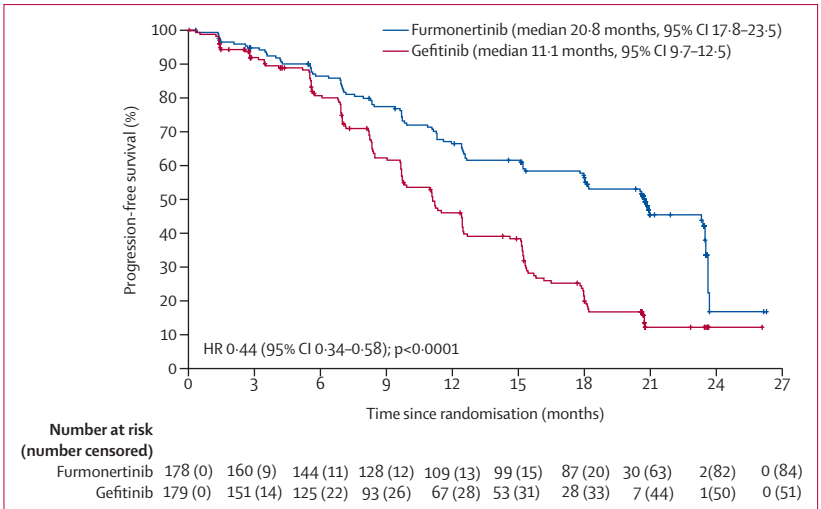

Figure 2: **IRC-assessed progression-free survival in the full analysis set**
HR=hazard ratio. IRC=independent review centre.

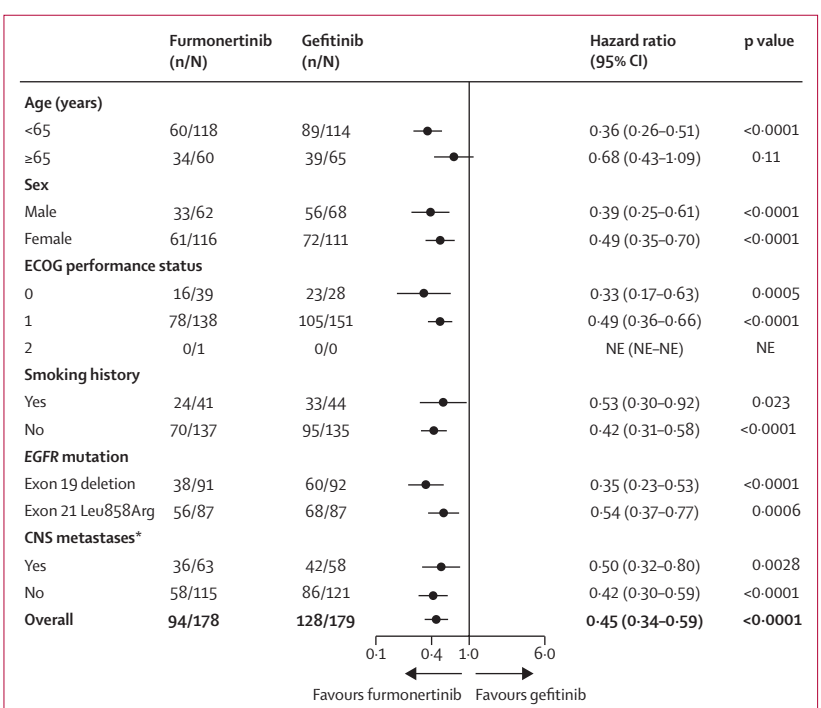

Figure 3: **Subgroup analysis of IRC-assessed progression-free survival in the full analysis set**
Hazard ratios and 95% CIs were calculated by use of an unstratified Cox proportional hazard model. p values were calculated with an unstratified log-rank test. ECOG=Eastern Cooperative Oncology Group. IRC=independent review centre. NE=not estimable. *CNS metastases were identified from baseline data for the CNS lesion site and medical history, including previous surgery or radiotherapy for CNS lesions, all of which were reported by investigators.

21·0 months (18·0–23·5) in the gefitinib group. Based on IRC assessment of the full analysis set, 94 (53%) patients had progressed or died in the furmonertinib group and 128 (72%) patients had progressed or died in the gefitinib group. Median IRC-assessed progression-free survival was 20·8 months (95% CI 17·8–23·5) in the furmonertinib group and 11·1 months (9·7–12·5) in the gefitinib group (HR 0·44, 95% CI 0·34–0·58; p<0·0001; figure 2). In the per-protocol set, IRC-assessed progression-free survival was similar to that in the full analysis set (appendix 2 p 31). Investigator-assessed progression-free survival was also significantly longer in the furmonertinib group than in the gefitinib group in the full analysis set (appendix 2 pp 10, 32).

In subgroup analyses, furmonertinib was associated with longer progression-free survival compared with gefitinib in most prespecified subgroups, including in patients with CNS metastases (HR 0·50, 95% CI 0·32–0·80; p=0·0028; figure 3). Kaplan–Meier curves for progression-free survival, as assessed by investigators and the IRC, in patients with and without CNS metastases can be found in appendix 2 (pp 33–34). In patients with CNS metastases, median investigator-assessed progression-free survival was 18·0 months (95% CI 12·4–23·3) with furmonertinib and 12·4 months (8·3–15·1) with gefitinib (appendix 2 p 33). In a post-hoc analysis of patients with concurrent *EGFR* exon 20 insertion mutations, IRC-assessed progression-free survival was 9·7 months and 20·6 months for two patients treated with furmonertinib and 3·4 months and 5·6 months for two patients treated with gefitinib (appendix 2 p 17). According to IRC assessment, 158 (89%, 95% CI 83 to 93) patients had an objective response in the furmonertinib group and 151 (84%, 78 to 89) patients had an objective response in the gefitinib group (OR 1·50, 95% CI 0·80 to 2·83; p=0·21; appendix 2 p 9). Among patients with an objective response, 77 (49%) of 158 in the furmonertinib group progressed or died and 107 (71%) of 151 in the gefitinib group progressed or died. IRC-assessed duration of response was significantly longer with furmonertinib than with gefitinib (19·7 months *vs* 11·0 months; HR 0·39, 95% CI 0·29 to 0·52; p<0·0001, appendix 2 p 35). The IRC-assessed rate of disease control was 96% (95% CI 91 to 98) with furmonertinib and 93% (89–97) with gefitinib (OR 1·56, 95% CI 0·61 to 3·98; p=0·36; appendix 2 p 9). The IRC-assessed median best percentage change in target lesion size (depth of response) from baseline was –61·1% (IQR –70·1 to –43·9) with furmonertinib and –55·9% (–69·4 to –41·2) with gefitinib (p=0·085). A waterfall plot for best percentage change in target lesion size is shown in appendix 2 (p 36). At data cutoff (Sept 15, 2021), 87 (49%) patients in the furmonertinib group and 125 (70%) patients in the gefitinib group had disease progression according to the IRC. Furmonertinib was associated with longer IRC-assessed time to progression than was gefitinib (HR 0·41, 95% CI 0·31–0·55; p<0·0001; appendix 2 p 37). Investigator-assessed outcome data for objective response rate, disease control rate, duration of response, depth of response, and time to progression were consistent with the results of the IRC assessment (appendix 2 p 10).

At the final analysis of progression-free survival, due to the short follow-up period, median overall survival was

not reached in either treatment group (appendix 2 p 38). 115 (32%) of 357 patients died, of whom 56 (31%) were in the furmonertinib group and 59 (33%) were in the gefitinib group. Thus, the maturity of overall survival was 32% (31% in the furmonertinib group and 33% in the gefitinib group) and did not meet the maturity requirement of the final overall survival analysis (60%). The HR for overall survival was 0·94 (95% CI 0·65–1·36; p=0·74; appendix 2 p 38). Mature overall survival results will be reported in future publications.

As for patient-reported outcomes on the EORTC QLQ-C30 and EORTC QLQ-L13 scales, furmonertinib resulted in significantly better overall changes from baseline to treatment cessation compared with gefitinib in physical function, nausea and vomiting, appetite loss, diarrhoea, alopecia, and pain in other parts of the body (appendix 2 pp 14–15). No clinically relevant difference in overall changes from baseline (≥10 points) between the furmonertinib group and the gefitinib group was found (appendix 2 pp 14–15). In either group, clinically relevant overall changes from baseline were seen in cough (–15·08 with furmonertinib and –16·28 with gefitinib) and chest pain (–10·39 with furmonertinib; appendix 2 pp 14–15).

By the cutoff date of Sept 15, 2021, the median duration of exposure to the study drug was 18·3 months (IQR 8·7–21·0) in the furmonertinib group and 11·2 months (5·6–18·1) in the gefitinib group. 178 (100%) of 178 patients who received furmonertinib and 177 (99%) of 179 patients who received gefitinib reported at least one treatment-emergent adverse event (appendix 2 p 11). According to investigators' assessments, 160 (90%) patients in the furmonertinib group and 170 (95%) patients in the gefitinib group had treatment-related adverse events (appendix 2 p 11). Elevated concentrations of aminotransferase enzymes was the most common treatment-related adverse event in both groups (table 2). Adverse events relevant to wild-type *EGFR* were also observed (31 [17%] of 178 had rash and 48 [27%] had diarrhoea in the furmonertinib group; 72 [40%] of 179 had rash and 64 [36%] had diarrhoea in the gefitinib group; table 2). Treatment-related QTc interval prolongation was reported in 16 (9%) of 178 patients in the furmonertinib group and in 12 (7%) of 179 patients in the gefitinib group. Decreases in left ventricular ejection fraction were not reported in either group. Interstitial lung disease was recorded in one patient in each group (grade 1 with furmonertinib and grade 2 with gefitinib).

Treatment-related adverse events of a grade 3 or more were reported in 20 (11%) patients in the furmonertinib group and in 32 (18%) patients in the gefitinib group (appendix 2 p 11). Treatment-related adverse events at grade 3 or more that occurred at a frequency of 2% or greater were ECG QTc interval prolongation and diarrhoea in the furmonertinib group and elevated alanine aminotransferase, elevated aspartate aminotransferase, ECG QTc interval prolongation, and anaemia in the gefitinib group (table 2). Treatment-related serious adverse events were recorded in ten (6%) patients who received furmonertinib and in 11 (6%) patients who received gefitinib. In both groups, the most common treatment-related serious adverse events were elevated alanine aminotransferase (four [2%] patients in the furmonertinib group; three [2%] patients in the gefitinib group) and elevated aspartate aminotransferase (three [2%] patients in the furmonertinib group; four [2%] patients in the gefitinib group; appendix 2 pp 11–12). Ten (6%) patients in the furmonertinib group and three (2%) patients in the gefitinib group died due to adverse events, which were

	Furmonertinib (n=178)			Gefitinib (n=179)		
	Grades 1–2	Grades 3–4	Grade 5	Grades 1–2	Grades 3–4	Grade 5
Elevated alanine aminotransferase	49 (28%)	2 (1%)	0	96 (54%)	13 (7%)	0
Diarrhoea	44 (25%)	4 (2%)	0	64 (36%)	0	0
Elevated aspartate aminotransferase	45 (25%)	2 (1%)	0	88 (49%)	11 (6%)	0
Rash	31 (17%)	0	0	70 (39%)	2 (1%)	0
Decreased white blood cell count	27 (15%)	0	0	20 (11%)	0	0
Oral ulcer	21 (12%)	0	0	16 (9%)	0	0
Decreased neutrophil count	18 (10%)	1 (1%)	0	11 (6%)	0	0
Anaemia	15 (8%)	1 (1%)	0	12 (7%)	3 (2%)	0
Decreased platelet count	14 (8%)	2 (1%)	0	9 (5%)	0	0
ECG QTc prolongation	10 (6%)	6 (3%)	0	8 (4%)	4 (2%)	0
Increased γ-glutamyltransferase	11 (6%)	2 (1%)	0	21 (12%)	1 (1%)	0
Decreased lymphocyte count	10 (6%)	1 (1%)	0	5 (3%)	0	0
Increased blood creatine phosphokinase	7 (4%)	0	0	8 (4%)	1 (1%)	0
Abdominal pain	5 (3%)	0	0	2 (1%)	1 (1%)	0
Weight loss	4 (2%)	0	0	1 (1%)	2 (1%)	0
Increased bilirubin	3 (2%)	1 (1%)	0	31 (17%)	1 (1%)	0
Increased conjugated bilirubin	2 (1%)	0	0	6 (3%)	1 (1%)	0
Paronychia	1 (1%)	1 (1%)	0	11 (6%)	0	0
Cerebral infarction	1 (1%)	1 (1%)	0	0	0	0
Chest discomfort	1 (1%)	0	0	0	1 (1%)	0
Abnormal liver function	0	1 (1%)	0	2 (1%)	2 (1%)	0
Decreased fibrinogen	0	1 (1%)	0	0	0	0
Obstructive pancreatitis	0	1 (1%)	0	0	0	0
Atrial fibrillation	0	1 (1%)	0	0	0	0
Cholecystitis	0	1 (1%)	0	0	0	0
Cholelithiasis	0	1 (1%)	0	0	0	0
Arrhythmia	0	0	0	2 (1%)	1 (1%)	0
Intestinal obstruction	0	0	0	0	1 (1%)	0
Bleeding from gastric ulcer	0	0	0	0	1 (1%)	0
Furuncle	0	0	0	0	1 (1%)	0

Data are n (%). Treatment-related adverse events were assessed by investigators. Grade 1–2 treatment-related adverse events that occurred in at least 10% of participants in either group and all grade 3–5 treatment-related adverse events are reported. ECG=electrocardiogram. QTc=heart-rate corrected QT.

Table 2: **Treatment-related adverse events in the full analysis set**

all judged to be possibly unrelated to study treatment by investigators (appendix 2 p 13).

Dose reductions of furmonertinib or furmonertinib-matched placebo due to treatment-emergent adverse events were recorded in five (3%) patients in the furmonertinib group and in six (3%) patients in the gefitinib group (appendix 2 p 11). Six (3%) patients who received furmonertinib permanently discontinued the study drug due to treatment-related adverse events, which were elevated alanine aminotransferase plus elevated aspartate aminotransferase (n=1), elevated bilirubin (n=1), decreased platelet count (n=1), ECG QTc interval prolongation (n=1), interstitial lung disease (n=1), and decreased visual acuity (n=1). In the gefitinib group, four (2%) patients permanently discontinued treatment due to treatment-related adverse events, which were elevated alanine aminotransferase plus elevated aspartate aminotransferase (n=1), interstitial lung disease (n=1), weight loss (n=1), and abnormal liver function (n=1). At data cutoff, 77 (43%) of 179 patients in the gefitinib group and 22 (12%) of 178 patients in the furmonertinib group had received subsequent post-progression third-generation EGFR tyrosine-kinase inhibitors (appendix 2 p 16).

Discussion

In this study, first-line furmonertinib showed significant benefits in progression-free survival compared with first-line gefitinib and an acceptable safety profile without new signals in patients with locally advanced or metastatic *EGFR*-mutated NSCLC. Other than the single ethnicity of participants, the demographic and clinical characteristics of patients in the FURLONG study were similar to those of patients in other studies.[16,17] The progression-free survival results favouring furmonertinib (20·8 months) versus gefitinib (11·1 months) in the FURLONG study are also similar to those from the FLAURA study[16] (18·9 months [95% CI 15·2–21·4] with osimertinib *vs* 10·2 months [9·6–11·1] with gefitinib or erlotinib; HR 0·46 [95% CI 0·37–0·57]; p<0·001) and the AENEAS study[17] (19·3 months [17·8–20·8] with aumolertinib [formerly named almonertinib] *vs* 9·9 months [8·3–12·6] with gefitinib; HR 0·463 [0·359–0·596]; p<0·0001), both of which investigated a third-generation EGFR tyrosine-kinase inhibitor versus one or two first-generation EGFR tyrosine-kinase inhibitors.

CNS metastases have been associated with worse prognosis in patients with NSCLC.[16] In the FURLONG study, 35% of patients in the furmonertinib group and 32% of patients in the gefitinib group had baseline CNS metastases. In patients with CNS metastases, median investigator-assessed progression-free survival was 18·0 months with furmonertinib and 12·4 months with gefitinib, and furmonertinib reduced the risk of progression or death by 50% compared with gefitinib, which shows the CNS activity of furmonertinib as a first-line treatment.

For this analysis, the overall survival benefit of furmonertinib versus gefitinib could not be concluded because the overall maturity was 32% (31% in the furmonertinib group and 33% in the gefitinib group) and the HR for overall survival was 0·94 (95% CI 0·65–1·36). In the FLAURA study, the overall survival benefit of osimertinib against gefitinib or erlotinib was significant (HR 0·80 [95% CI 0·64–1·00]; p=0·046) at 58% maturity.[9] Due to immature survival data, long-term follow-up of survival is needed to better characterise the effect of furmonertinib on overall survival. Another possible reason for the overall survival results of this study might be the higher rate of post-progression use of third-generation EGFR tyrosine-kinase inhibitors in the gefitinib group compared with the furmonertinib group. The crossover rate in the control group of the FLAURA study was 31% in the final overall survival report.[9] However, in the FURLONG study, 43% of patients in the gefitinib group versus 12% of patients in furmonertinib group had received subsequent third-generation EGFR tyrosine-kinase inhibitors at data cutoff. In addition, the overall survival benefit of osimertinib over gefitinib or erlotinib in the Asian subgroup in the FLAURA study was smaller, with a HR of 1·00, than that in the whole study population.[9] The final overall survival results of the FURLONG study will be reported when the data are mature.

The safety profiles of furmonertinib and gefitinib reported in the FURLONG study are in line with the results of previous studies.[3,11,12] Wild-type *EGFR*-related adverse events are commonly reported in studies of EGFR tyrosine-kinase inhibitors, with a prevalence of around 59–81% for rash and 46–88% for diarrhoea.[3,16,18,19] The relatively low prevalences of treatment-related rash (17%) and diarrhoea (27%) in the furmonertinib group in our study is consistent with previous results.[11,12] Moreover, we found that furmonertinib resulted in significantly better overall changes from baseline compared with gefitinib in several patient-reported outcome scores for multiple gastrointestinal items, including diarrhoea, nausea and vomiting, and appetite loss. All these data together further show the high selectivity of furmonertinib against mutated *EGFR*.

A previous phase 2b study has revealed the mechanism of furmonertinib in patients with NSCLC with the *EGFR* Thr790Met mutation.[12] The most common *EGFR*-dependent resistance mechanism was a tertiary *EGFR* mutation at Cys797 (9%). *EGFR*-independent resistance mechanisms included *HER2* (also known as *ERBB2*) amplification (9%), *MET* amplification (6%), and *KRAS* mutation (6%). The Cys797Ser mutation in *EGFR* was also the most common *EGFR*-dependent resistance mechanism of osimertinib and aumolertinib.[20,21] Among non-*EGFR* alterations, *MET* amplification was the most frequent (15%) with osimertinib[20] and *PIK3CA* mutations (eg, those resulting in His1047Ter, Glu545Lys, and Glu542Lys) were the most common (23%) with aumolertinib.[21] In our study, blood samples are still being

collected and we will report resistance mechanisms when the biomarker analyses are completed.

In the ongoing FAVOUR study (NCT04858958), high-dose (240 mg/day) furmonertinib showed a preliminary objective response rate of 60% (six of ten participants) in treatment-naive patients with NSCLC and *EGFR* exon 20 insertions.[22] Four patients with such mutations were enrolled in the FURLONG study and their signals were also positive in a post-hoc analysis; further data are expected. Because of promising efficacy and safety data, furmonertinib is also being investigated in a phase 3 study as adjuvant therapy in patients with resected, stage II–IIIA NSCLC (FORWARD; NCT04853342) and in studies as perioperative or adjuvant therapy in other populations with early-stage NSCLC (NCT04965831, NCT05079022, and NCT05165355). Trials investigating furmonertinib combined with anlotinib in patients with advanced *EGFR*-mutated NSCLC (NCT04895930) and furmonertinib plus radiotherapy in patients with NSCLC oligoprogression (NCT04970693) are also ongoing. A licensing agreement has been reached between Shanghai Allist Pharmaceuticals (Shanghai, China) and ArriVent Biopharma (Newtown Square, PA, USA) and the development of furmonertinib outside of China will be initiated by ArriVent Biopharma.[23]

Besides osimertinib and furmonertinib, some other third-generation EGFR tyrosine-kinase inhibitors are also in active clinical development, including aumolertinib, lazertinib, rezivertinib, ASK120067, D-0316, and SH-1028.[24] To our knowledge, among the aforementioned drugs, data have been presented for only osimertinib and aumolertinib as first-line treatments.[16,17] Based on reported data, the safety profiles of different third-generation EGFR tyrosine-kinase inhibitors might be different.[25] The frequencies of diarrhoea, rash, or acne were more than 50% with osimertinib,[16] but no more than 30% with furmonertinib or aumolertinib as first-line therapies.[17] The prevalence of elevated creatine phosphokinase was high with aumolertinib (76 [36%] of 214 patients for all severity grades; 15 [7%] patients for grade 3 or higher).[17] Although there are no data for first-line D-0316, data for second-line D-0316 have shown that thrombocytopenia was the most common adverse event (166 [57%] of 290 patients for all grades and 34 [12%] patients for grade 3 or higher in the 75–100 mg group).[26] These differences in safety profiles could be due to the different structures of the parent drugs and their metabolites.

Our study has several strengths. First, this study is a rigorously designed, double-blind, double-dummy trial with a primary outcome of progression-free survival by a blinded IRC. Second, patients with asymptomatic CNS metastases could be enrolled and diagnosis of CNS metastases was based on mandatory brain scanning at baseline. Finally, the stratification factors of *EGFR* mutation type and CNS metastases status, together with other well balanced baseline characteristics, enhance the comparability between groups.

One limitation of this study is the single ethnicity of its participants. The efficacy and safety of furmonertinib remain to be explored in other populations. A further limitation is the exclusion of other first-generation and second-generation EGFR tyrosine-kinase inhibitors as comparators. However, gefitinib was the most widely used EGFR tyrosine-kinase inhibitor in mainland China when FURLONG was designed. At the time the study was designed, afatinib was also approved for first-line treatment of NSCLC positive for *EGFR* sensitizing mutations. However, there was no solid evidence that afatinib had superior efficacy over first-generation EGFR tyrosine-kinase inhibitors as first-line therapy for NSCLC and afatinib was associated with a higher prevalence of grade 3–4 adverse events compared with gefitinib.[26] Similarly, gefitinib was also the only investigational comparator in the FLAURA China cohort.[15]

In conclusion, furmonertinib had significantly superior efficacy compared with gefitinib as first-line therapy in patients with locally advanced or metastatic *EGFR* mutation-positive NSCLC, along with an acceptable safety profile without new signals. Furmonertinib should be considered as a new treatment option for treatment-naive Chinese patients with *EGFR*-mutated NSCLC.

Contributors

YS was the leading principal investigator and designed the trial with the sponsor. YS, GC, XW, YLiu, LW, YH, CL, SZhu, XZ, YLi, JL, LC, YC, HZ, SZha, AZ, JC, JF, and NY contributed to patient recruitment and data acquisition. YS (academic), FL, and CG accessed and verified the data. FL, YJ, and CG were involved in data analysis and data interpretation. CG did the manuscript writing. All authors reviewed and approved the final version of the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

FL, YJ, and CG are employees and shareholders of Shanghai Allist Pharmaceuticals. All other authors declare no competing interests.

Data sharing

The data generated and analysed in this study are on file with Shanghai Allist Pharmaceuticals and are not publicly available according to company policy.

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